

HAILUOTO MARJANIEMI, Finland

WMO: 028730

Lat: 65.0400N

Lon: 24.5615E

Elev: 7

StdP: 101.24

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.9	-20.8	-26.4	0.3	-23.9	-22.8	0.5	-20.7	19.0	-1.5	17.6	-1.4	3.7	130	0.872

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.6	23.6	18.9	21.7	17.8	20.1	17.0	19.8	22.5	18.5	20.9	17.4	19.6	4.8	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.6	21.1	17.6	12.6	20.0	16.4	11.7	18.8	56.6	22.4	52.5	20.8	49.1	19.9	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.2	15.3	13.6	-25.2	26.2	4.4	2.8	-28.4	28.3	-30.9	30.0	-33.4	31.5	-36.6	33.6		
(5)			-25.4	20.7	4.4	2.0	-28.5	22.1	-31.1	23.3	-33.5	24.4	-36.7	25.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.7	-7.1	-7.9	-5.2	0.2	6.9	12.2	16.4	15.2	10.8	4.8	0.4	-2.8	
	DBStd	9.33	6.35	6.76	5.16	3.07	4.14	3.15	2.98	2.69	2.75	3.30	3.86	5.26	
	HDD10.0	2789	528	501	472	293	114	12	0	1	23	162	288	396	
	HDD18.3	5349	787	734	731	543	354	186	77	102	225	419	538	654	
	CDD10.0	507	0	0	0	0	19	78	198	163	48	1	0	0	
	CDD18.3	26	0	0	0	0	1	2	17	6	0	0	0	0	
	CDH23.3	76	0	0	0	0	6	8	49	13	0	0	0	0	
	CDH26.7	12	0	0	0	0	1	1	9	1	0	0	0	0	
Wind	WSAvg	6.7	6.8	6.6	6.9	6.2	6.2	6.1	5.6	6.1	7.1	7.6	7.7	7.6	
Precipitation	PrecAvg	520	38	33	30	28	38	42	63	55	49	50	49	43	
	PrecMax	694	73	69	81	78	84	102	119	117	90	97	84	86	
	PrecMin	371	14	14	5	8	6	2	3	4	11	7	10	12	
	PrecStd	85	17	17	17	16	18	27	30	26	24	21	19	20	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.6	2.9	3.9	12.3	23.1	24.2	27.9	25.0	19.4	12.1	8.5	4.7	
		MCWB	1.7	1.8	1.8	7.6	14.9	17.0	20.6	19.3	15.4	10.7	7.3	3.9	
	2%	DB	1.7	1.9	2.6	7.7	18.4	20.7	24.2	22.3	16.6	11.1	7.5	3.9	
		MCWB	1.2	1.2	1.3	4.4	12.6	15.8	19.8	18.2	14.2	9.9	6.8	3.1	
	5%	DB	1.1	1.1	1.8	5.2	15.1	18.2	22.4	20.5	15.3	10.0	6.2	3.2	
		MCWB	0.6	0.6	0.9	2.9	11.2	14.6	18.7	17.3	13.6	9.0	5.6	2.5	
	10%	DB	0.4	0.3	1.1	3.6	12.2	16.5	20.9	19.1	14.4	9.1	5.2	2.4	
		MCWB	0.0	-0.1	0.4	2.1	9.1	13.6	17.9	16.6	12.9	8.1	4.6	1.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.1	2.1	2.5	8.1	15.6	18.2	21.9	20.7	15.9	11.2	7.7	4.3
			MCDB	2.3	2.6	3.6	10.4	21.3	21.7	24.2	23.6	17.8	11.8	8.1	4.4
		2%	WB	1.4	1.4	1.7	4.8	13.2	16.7	20.5	19.1	14.7	10.1	6.9	3.4
			MCDB	1.6	1.8	2.3	7.3	17.9	19.5	23.2	21.3	16.2	10.9	7.5	3.7
5%		WB	0.7	0.7	1.1	3.3	11.4	15.2	19.3	18.0	13.9	9.2	5.8	2.6	
		MCDB	1.0	1.1	1.7	4.6	14.3	17.6	21.8	19.7	15.0	9.9	6.2	3.0	
10%		WB	0.2	0.0	0.5	2.3	9.7	14.1	18.2	16.9	13.2	8.3	4.7	1.9	
		MCDB	0.5	0.4	1.0	3.3	11.9	16.2	20.7	18.7	14.3	9.0	5.1	2.4	
Mean Daily Temperature Range		5% DB	MDBR	5.1	5.5	6.0	5.1	5.3	4.6	4.6	4.5	3.7	3.2	3.3	3.8
			MCDBR	3.5	3.9	4.4	7.2	9.7	7.9	7.4	6.9	5.1	3.3	3.2	3.1
	MCWBR		3.5	3.7	3.7	5.0	5.5	4.6	4.2	4.2	3.4	2.9	3.1	2.8	
	5% WB	MCDDBR	3.7	3.8	4.0	6.4	8.5	6.6	6.3	5.5	4.7	3.1	3.2	3.0	
		MCWBR	3.7	3.6	3.5	4.7	5.5	4.3	4.1	4.1	3.4	2.9	3.2	2.9	
Clear-Sky Solar Irradiance	taub	0.209	0.276	0.304	0.341	0.346	0.350	0.369	0.354	0.331	0.308	0.343	0.353		
	taud	1.950	2.092	2.133	2.173	2.407	2.488	2.455	2.489	2.529	2.354	2.545	2.589		
	Ebn at Noon	292	596	759	820	861	866	834	813	756	577	268	110		
	Edh at Noon	23	60	91	112	99	94	94	83	65	48	21	12		
All-Sky Solar Radiation	RadAvg	0.10	0.61	2.09	3.79	4.96	5.56	5.14	3.80	2.08	0.75	0.16	0.03		
	RadStd	0.01	0.12	0.21	0.30	0.47	0.44	0.46	0.36	0.21	0.12	0.03	0.00		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page